

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fourth Semester B.E. (CSE/ISE/AIIML/CCE) (Credit System) Degree Examinations****Make up / Supplementary Examinations - September 2022****19MA401/20CS401/20IS401/20AM401/20CC401 – LINEAR ALGEBRA AND PROBABILITY THEORY**

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – III** each and **One full** question from **Unit – II**.

2) Use of Normal Distribution Table is permitted.

Unit – I		Marks	CO*
1.	a) Define linearly dependent and linearly independent set of vectors. Prove that a set of vectors is linearly dependent if and only if at least one of them is linear combination of the others.	7	1
	b) Find the coordinates of the vector (a, b, c) in the real vector space R^3 with respect to the basis $\{(1, 0, -1), (1, 1, 1), (1, 0, 0)\}$	6	1
	c) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$	7	2
2.	a) Define a basis and dimension for a vector space. If $\{u_1, u_2, \dots, u_n\}$ is a basis for a vector space V , then prove that any vector in V can be expressed as a unique linear combination of basis vectors.	7	1
	b) Check whether the set of vectors $\{(1, 2, -1), (1, -2, 1), (-3, 2, -1)\}$ is linearly dependent or linearly independent.	6	1
	c) Find trace (A) and spectrum of A where $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$	7	2
3.	a) Define a subspace of a vector space V . If G and H are subspaces of a vector space V prove that $G \cap H$ is also a subspace of V .	7	1
	b) Find the gradient and Hessian of $f(x, y, z) = 3x^3 + 2y^2 + 3z^2$	6	1
	c) Find the value of K if the vector $(1, -2, K)$ in R^3 is a linear combination of the vectors $(3, 0, -2)$ and $(2, -1, -5)$.	7	1
Unit – II			
4.	a) Show that $T: V_2(R) \rightarrow V_2(R)$ defined by $T(x, y) = (3x + 2y, 3x - 4y)$ is a linear transformation.	7	3
	b) The linear transformation $T: V_2(R) \rightarrow V_2(R)$ is defined by $T(x_1, x_2) = (x_1 + x_2, 2x_1 - x_2)$. Find the matrix of the transformation relative to the bases $B = \{(1, 2), (-1, 1)\}$ and $B' = \{(1, 0), (0, 1)\}$	6	3
	c) Using Gram-Schmidt orthogonalization process find an orthonormal basis from the basis $\{(1, 0, 1), (1, 0, -1), (0, 3, 4)\}$ of R^3 .	7	3
5.	a) Find the matrix of the linear transformation $T: V_2(R) \rightarrow V_3(R)$ defined by $T(x, y) = (x + y, x, 3x - y)$ with respect to bases $\beta = \{(1, 1), (3, 1)\}, \beta' = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$	7	3

- b) The linear transformation $T: V_3(R) \rightarrow V_3(R)$ is defined by $T(x, y, z) = (x + y, x - y, 2x + z)$. Find the range space, null-space, rank and nullity of T and verify that rank of T + nullity of $T = \dim(V_3(R))$
- c) Find a least square solution of the inconsistent system

$$AX = B \text{ where } A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

6 3

7 3

Unit – III

6. a) In a bolt factory machines A, B and C manufacture respectively 25%, 35% and 40% of the total. Of their outputs 5, 4 and 2 percent are defective bolts. A bolt is chosen at random and is found to be defective. What is the probability that it was manufactured by machine B?

7 4

- b) A random variable X has the following probability function.

X	0	1	2	3	4	5	6	7
$p(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

- i) Find K ii) Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(3 < X \leq 6)$

6 4

- c) Two random variables X and Y have the following joint p.d.f.

$Y \backslash X$	-1	0	1
-1	0.1	0.1	0
0	0	0.3	0.1
1	0.1	0.2	0.1

Find

- i) Marginal pdf's of X and Y
 ii) $P[X \leq 0, Y \geq 0]$ iii) $P[X + Y \leq 0]$

7 4

7. a) State and prove Baye's Theorem.
 b) A continuous random variable X has the pdf.

$$f(x) = \begin{cases} 6x(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $E[X]$ and $V[X]$.

7 4

6 4

- c) A continuous random variable X has the density function

$$f(x) = \begin{cases} Kx^2, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

Find

- i) the value of K ii) $P(1 \leq X \leq 2)$ iii) $P(X \leq \frac{1}{2})$ iv) $P(X > \frac{1}{2})$

7 4

8. a) Find the mean and variance of exponential distribution.
 b) If the probability that a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals more than two will get a bad reaction.
 c) If X has a normal distribution with mean 30 and standard deviation 5, find
 i) $P(26 \leq X \leq 40)$ ii) $P(X \geq 45)$ iii) $P(|X - 30| > 5)$

7 5

6 5

7 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
